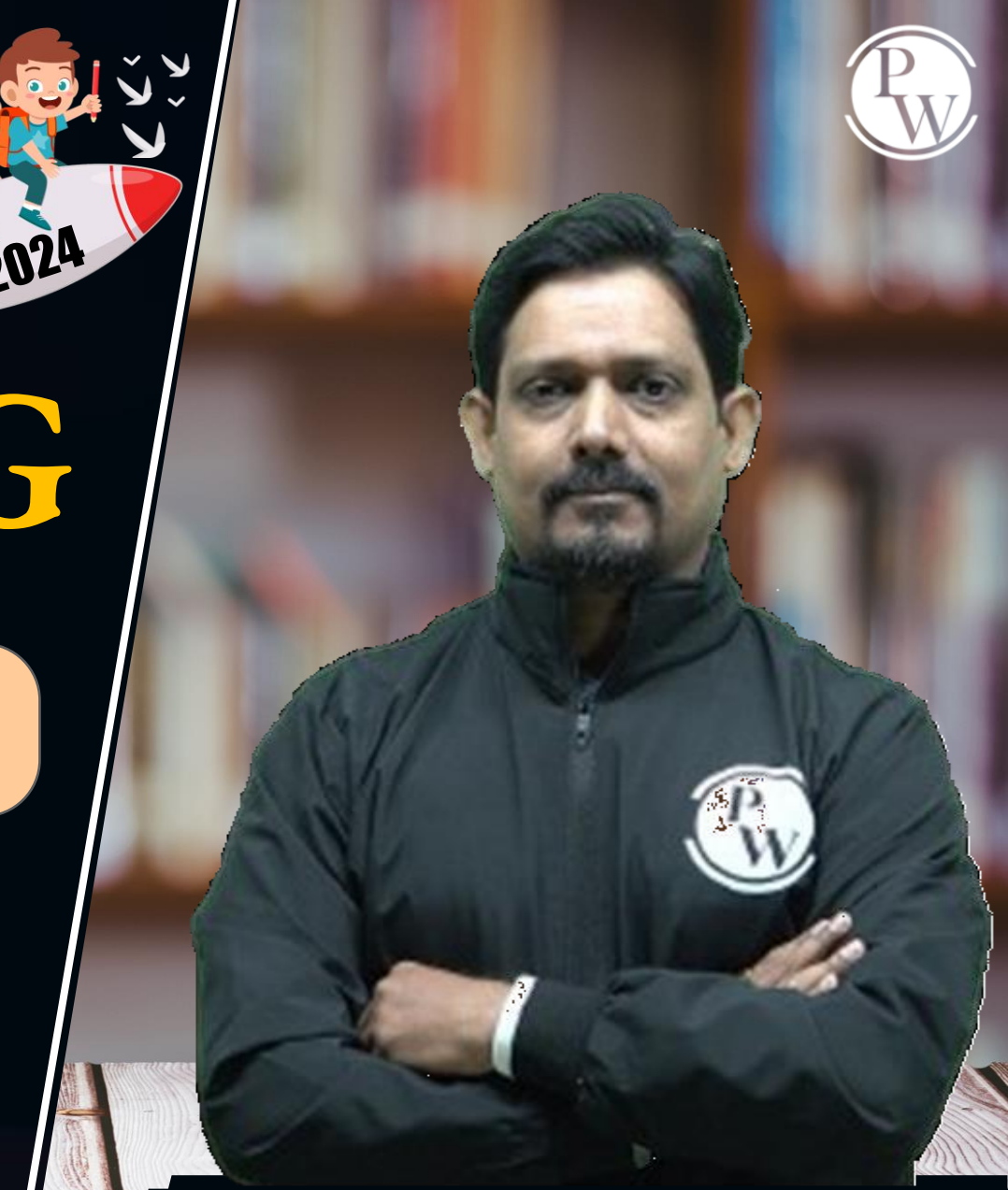
CS & IT ENGINEERING



Data Structure

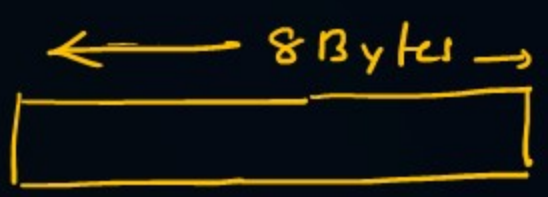
Structure & union

DPP-01 Discussion Notes



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[NAT]



#Q.
```
#include <stdio.h> union u{
    int a;
    char b;
    double d[2];
};
int main() {
    union u u1
     printf("%d",(int)sizeof(u1)); return 0;
}
```

int a; ✓ 2 Byte

char b; ✓ 1 Byte

double d[2]; ✓ 4B = 4 x 2B = 8B

Max (2, 1, 8) = 8 Bytes

8

Assume that objects of the type int, char and double occupy 2 bytes, 1 bytes and 4 bytes, respectively. The memory requirement for variable u1 is (in bytes).

[NAT]

#Q. Consider the following C declaration:

```
struct {
        long a[3];
        union {
                int y;
                float z;
        }u;
} s;
```

int 2B
float 4B
long 8B

long a[3] ⟶ 24B

union { int y; float z; } ⟶ max(2,4) = 4B

24B + 4B = 28 Bytes

Assume that objects of the type int, float and long occupy 2 bytes, 4 bytes and 8 bytes, respectively. The memory requirement for variable s is (in bytes).

28 Bytes

[MCQ]

#Q.

```
#include <stdio.h>
struct s{
char a, b;
};
void f(struct s *p){
 p->a+=2;
p->b-=1;
}
int main(){
struct s s1, *q;
        s1.a='A';
        s1.b='C';
        q=&s1;
        f(q);
        printf("%c\t%c",s1.a, s1.b);
        return 0;
}
```

s_1 *q structure pointer

[A]

structure

structure pointer

function

$97+2 = 99$ — 'C'

$99-1 = 98$ — 'B'

The output is

A. C B

B. A C

C. Compilation error

D. Garbage Value

[MCQ]

A, C [B]

#Q.
```
#include <stdio.h>
struct s{
        char a, b;
};
void f(struct s s1){
s1.a+=3;
s1.b-=1;
}
int main() {
 struct s s1; s1.a='A'; s1.b='C';
f(s1);
printf("%c\t%c",s1.a, s1.b); return 0;}
```

s1 → structure variable

f → Local variable

It will Not reflect

The output is

A. C B

B. A C

C. Compilation error

D. Garbage Value

[MCQ]

#Q.

```
#include <stdio.h>
 struct s{
char a, b;
};
void f(struct s s1){
s1.a+=32;
s1.b+=32;
}
void g(struct s *p){
static count=2;
p->a+=count++;
p->b+=++count;}
int main(){
struct s s1, s2;
s1.a='A'; s1.b='C';
s2.a='B'; s2.b='D';
f(s1);
for(int i=0;i<2;i++)
        g(&s2);
printf("%c\t%c",s1.a, s1.b);
printf("\t%c\t%c",s2.a, s2.b);
return 0;
}
```

Structure variable types

structure pointer

[C]

The output is

A. a c B D ✗

B. A C B D

C. A C H N ✓

D. a c B N ✗

[MCQ]

#Q.

```
#include <stdio.h>
 struct s{
char a, b;
};
void f(struct s s1){
s1.a+=32;
s1.b+=32;
}
void g(struct s *p){
static count=2;
p->a+=count++;
p->b+=++count;}
```

B, D
↓ ↓
D H
↓
H N

E F G H, I, J, K L, M, N

L=0 — D H

L=1 — H N

```
int main(){
struct s s1, s2;
s1.a='A'; s1.b='C';
s2.a='B'; s2.b='D';
f(s1);
for(int i=0;i<2;i++)
        g(&s2);
printf("%c\t%c",s1.a, s1.b);
printf("\t%c\t%c",s2.a, s2.b);
return 0;
}
```

static



[MCQ]

```
#include <stdio.h>
struct days{
char *q;
}s[]={"Sunday", "Monday", "Tuesday",
"Wednesday", "Thursday", "Friday", "Saturday"};
int main() {
    struct days *p=s; p=p+3;
    printf("%c", *p++->q);
    printf("%c", *++p->q); p=p-2;
    printf("%s",p->q); return 0;
  }
```

[D]

The output printed is

A WhWednesday

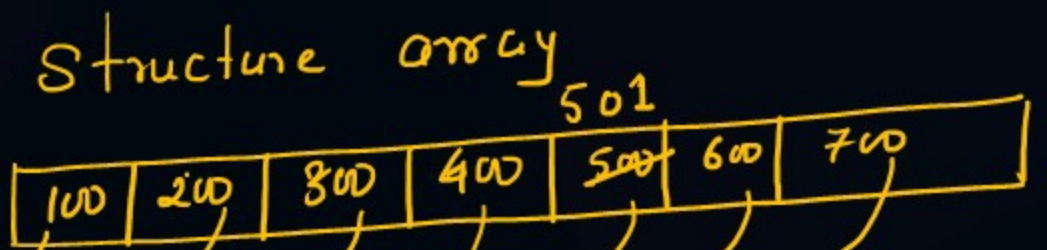
B WTTuesday

C WTWednesday

D WhTuesday

[MCQ]

```
#include <stdio.h>
struct days{
char *q;
}s[]={"Sunday", "Monday", "Tuesday",
"Wednesday", "Thursday", "Friday", "Saturday"};
int main() {
    struct days *p=s; p=p+3;
    printf("%c", *p++->q);
    printf("%c", *++p->q); p=p-2;
    printf("%s",p->q); return 0;
  }
```



W

WhTuesday

400 ↑

p | 400 | 500 | 300

* p++ → q

*(p → q)

(p → q)

[MSQ]

Which of the following statements are INCORRECT?

A. Functions cannot be defined inside the structure [T]

B. Structure variable of the same structure type can be defined inside a structure. F

C. A function may not contain a structure defined in it. [F]

D. Existing structure cannot be contained in another structure. [F]

[B] [C] [D]

[NAT]

No. of Lines of error

#Q.
```
#include<stdio.h>
#include<string.h>
struct t {
char sname[20];
};
int main () {
struct t t1, t2;
strcpy(t1.sname, "GATEWallah"); //line a
t2.sname="GATE2023"; //line b
printf("%s", t1.sname); //line c
printf("%s", t2.sname); //line d
return 0; }
```

character array

The number of lines with error among lines a,b,c,d are 2

Assignment Not possible

GATEWallah

Error

THANK - YOU

